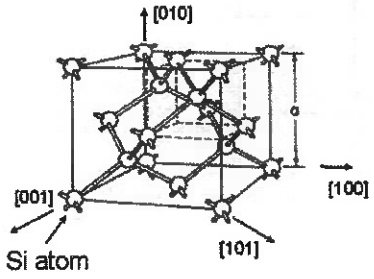
Solutions 1:

a) Inspecting the unit cell:



Total number of atoms in the unit cell $N_t = 18$. However, in the crystal a group of only $n = 8$ atoms repeats, the remaining 10 atoms lie in the next unit cells in the x, y and z directions (2 marks until this point). Volume of unit cell $V = a^3 = 1.57 \times 10^{-28}$ m^3. Therefore, atomic concentration $D = n/V = 8/1.57 \times 10^{-28} = 5 \times 10^{28}$ atoms/m^3 (remaining 2 marks).

[Marks: 4]

b) In the Deal-Grove equation, τ corresponds to the 'time' corresponding to the initial oxide thickness $X_i = 10$ nm. Working in Angstrom, we first calculate τ:

$$\tau = (X_i^2 + AX_i)/B = (100^2 + 0.3 \times 10^4 \times 100) / 2.9 \times 10^4 = 10.7 \text{ min.}$$

(2 marks)

Therefore, at $t = 50$ min., we have:

$$X^2 + AX = B(t + \tau)$$

$$\Rightarrow \quad X^2 + (0.3 \times 10^4)X = (2.9 \times 10^4)(50 + 10.7) = 1.76 \times 10^6$$

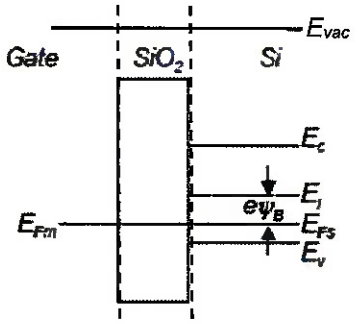
$$\Rightarrow \quad X = (-3000 \pm \sqrt{(3000^2 - 4(-1.76 \times 10^6)})/2$$

$$\Rightarrow \quad X = 50 \text{ nm}$$

Here we have used the positive root only (remaining 3 marks).

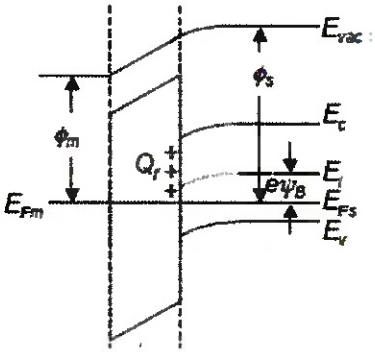
[5]

(c) (i) Band diagram for 'flat–band' conditions:



[3]

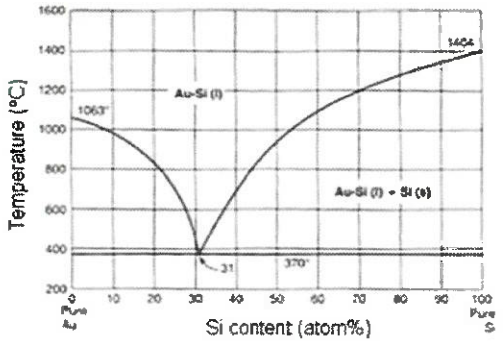
(ii) Band diagram with trapped charge $+Q_f$ at the gate oxide/Si interface, and work function in the gate metal $\phi_m < \phi_s$, the work function in the Si.



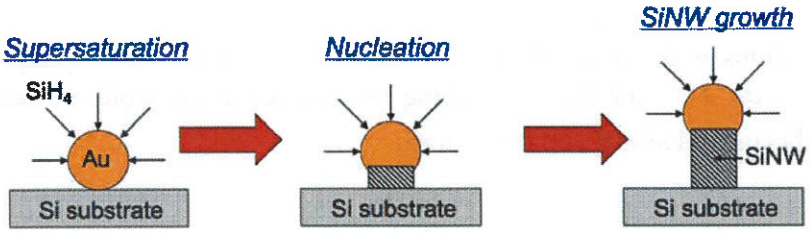
[3]

(d)

The vapour-liquid-solid (VLS) growth mechanism uses metal nanoparticles (typically Au or Fe) as a catalyst to grow Si nanowires. The wires may be grown by CVD, or PECVD, from SiH_4 gas, in the presence of Au nanoparticles on a substrate. The Au-Si phase diagram is shown below (this may be shown schematically, 2 marks for this):



The phase diagram predicts that Au nanoparticles would form a liquid Au-Si alloy at temperature $T > 363°C$ (Eutectic point). For this temperature range, during the CVD/PECVD process, super-saturation of Si in the nanoparticle alloy leads to precipitation of solid Si and the formation of a solid Si/liquid Si/Au-Si alloy interface. Subsequent growth of a 1-D Si nanowire occurs from this interface. The process is shown schematically in the figure below:



3 marks for discussion of this nature.

[5]

Answer, Question 1:

e) The generalised Hooke's law equations are:

$$E\varepsilon_x = \sigma_x - \upsilon\sigma_y - \upsilon\sigma_z$$
$$E\varepsilon_y = \sigma_y - \upsilon\sigma_z - \upsilon\sigma_x$$
$$E\varepsilon_z = \sigma_z - \upsilon\sigma_x - \upsilon\sigma_y$$

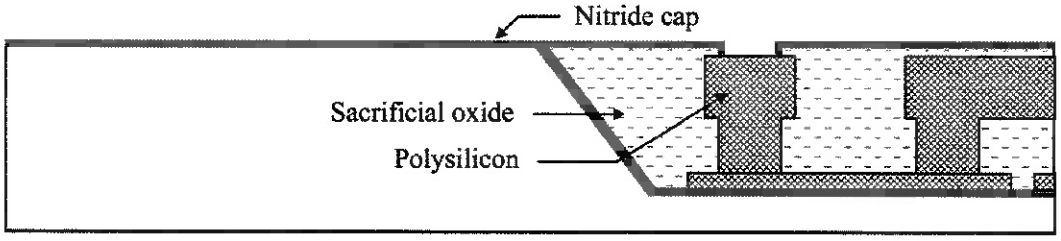
The biaxial modulus is the ratio of stress to strain when there is isotropic plane stress, for example if $\sigma_x = \sigma_y \neq 0$ and $\sigma_z = 0$. Taking this particular example, we see that the first two equations become identical and of the form:

$$E\varepsilon = \sigma(1-\upsilon)$$

The ratio of stress to strain, which is the biaxial modulus, is then $\tilde{E} = \dfrac{E}{1-\upsilon}$. [5]

f) Materials are: A – sacrificial oxide; B – polysilicon; C – CMOS metallisation, e.g. aluminium, D – silicon nitride passivation; CMOS dielectric, e.g. oxide.

Cross-section immediately before CMOS process:

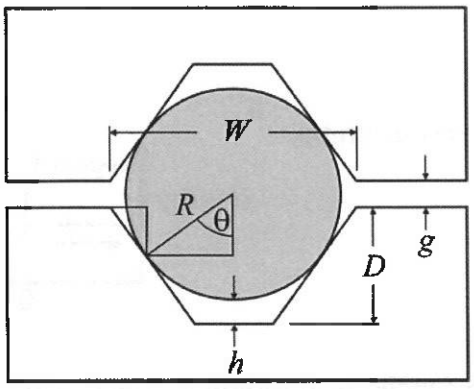


Advantages: (1) more freedom over MEMS deposition/annealing conditions in absence of CMOS circuitry; (2) standard CMOS foundry process can be used. [5]

g) From the diagram below, the side length W at the wafer surface needs to satisfy:

$$W/2 = R\sin\theta + (R\cos\theta - g/2)\cot\theta = (\sqrt{3}R - g/2)/\sqrt{2}$$

where we have used the fact that the angle between the V-groove sidewalls and the wafer surface is $\theta = \tan^{-1}\left(\sqrt{2}\right)$. With R = 250 μm and g = 200 μm, this gives W = 470.95 μm. The mask undercut is given by $\delta = \sqrt{6}D/S$ where D is the etch depth and S is the anisotropy. The etch depth is $D = R - g/2 + h$ = 175 μm, so δ = 1.07 μm and the required mask aperture size is 469.9 × 469.9 μm^2.



[5]

h) A grating mask of period d will generate diffracted beams at angles θ_n given by the Bragg condition:

$$\sin\theta_n = n\frac{\lambda}{d} \qquad n = 0, \pm 1, \pm 2 \ldots$$

The finest grating that will be resolved is the one for which the ±1 diffraction orders just pass through the projection lens. In this case we have $\theta_1 = \theta_m$ where θ_m is the maximum acceptance angle of the lens. But $\sin\theta_m = NA_o$ where NA_o is the numerical aperture on the object side, so for the finest resolvable grating we have:

$$d_{\min} = \frac{\lambda}{NA_o}$$

and the minimum resolvable half-pitch on the object side is:

$$R_o = \frac{d_{\min}}{2} = \frac{\lambda}{2NA_o}$$

The corresponding half-pitch on the image side is $R = R_o/m$, where m is the magnification, while the image side numerical aperture is $NA = mNA_o$, so we can recast the above formula for the image side as:

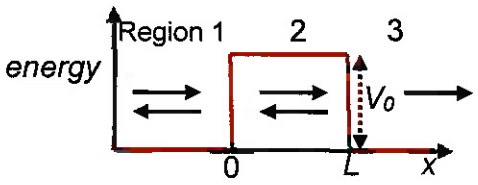
$$R = \frac{\lambda}{2NA}$$

[5]

Solution 2:

Here, the potential energy $V(x)$ given by:

$V(x) = V_0,\ 0 \le x \le L$
$V(x) = 0,\ x < 0,\ x > L$



(a) Space-dependent wave-functions in region 1, 2 and 3 are:

Region 1

$$\psi_1 = a_{1i}\exp(ik_1x) + a_{1r}\exp(-ik_1x) = \exp(ik_1x) + r\exp(-ik_1x)$$

where we assume $a_{1i} = 1$
and r = amplitude reflection coeff.

Region 2:

$$\psi_2 = a_{2i}\exp(ik_2x) + a_{2r}\exp(-ik_2x)$$

Region 3:

$$\psi_3 = a_{3i}\exp(ik_1x)$$

[Marks: 2 +2 + 2]

Wave-vectors in regions 1 and 2, for energy E, are as follows:

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}} \qquad k_2 = \sqrt{\frac{2m(E-V_0)}{\hbar^2}}$$

Region 3 is identical to region 1, therefore $k_3 = k_1$

[Marks: 2]

(b)

The boundary conditions at $x = 0$ are:

$$\psi_1 = \psi_2 \quad \text{and} \quad \frac{d\psi_1}{dx} = \frac{d\psi_2}{dx}$$

The boundary conditions at $x = L$ are:

$$\psi_2 = \psi_3 \quad \text{and} \quad \frac{d\psi_2}{dx} = \frac{d\psi_3}{dx}$$

[Marks: 4 + 4]

(c)

Boundary conditions:

At $x = 0$: $\psi_1 = \psi_2 \quad \Rightarrow \quad 1 + r = a_{2i} + a_{2r}$ (1)

$$\frac{d\psi_1}{dx} = \frac{d\psi_2}{dx} \quad \Rightarrow \quad k_1(1-r) = k_2(a_{2i} - a_{2r}) \qquad (2)$$

Similarly, at $x = L$: $a_{2i}e^{ik_2L} + a_{2r}e^{-ik_2L} = a_{3i}e^{ik_1L} = t$ (3)

$$k_2\left(a_{2i}e^{ik_2L} - a_{2r}e^{-ik_2L}\right) = k_1 t \qquad (4)$$

where t = amplitude transmission coeff.

Dividing Eq. 4 by k_2 and adding to Eq. 3 $\Rightarrow$

$$a_{2i} = \left(1 + \frac{k_1}{k_2}\right)\frac{te^{-ik_2L}}{2} \qquad (5)$$

[4 Marks]

Substituting Eq. 5 into Eq. 3 $\Rightarrow$

$$a_{2r} = \left(1 - \frac{k_1}{k_2}\right)\frac{te^{ik_2L}}{2} \qquad (6)$$

[4 Marks]

Dividing Eq. 2 by k_1 and adding to Eq. 1 $\Rightarrow$

$$2 = a_{2i}\left(1 + \frac{k_2}{k_1}\right) + a_{2r}\left(1 - \frac{k_2}{k_1}\right) \qquad (7)$$

Finally, substituting Eq. 5 and 6 into Eq. 7 $\Rightarrow$

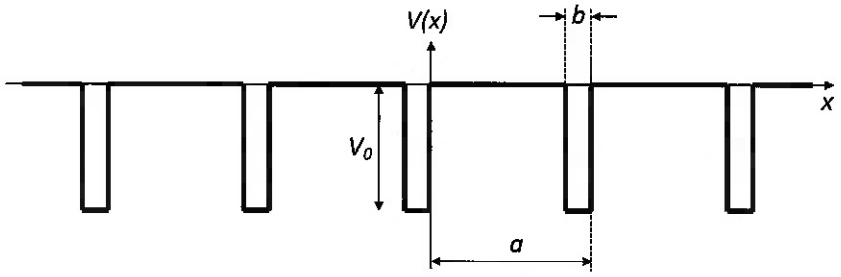
$$\boxed{t = \frac{4k_1k_2}{(k_1+k_2)^2 e^{-ik_2L} - (k_1-k_2)^2 e^{ik_2L}}} \qquad (8)$$

[6 Marks]

[Total Marks: 14]

Solution 3:

a)



(i) For region $0 < x < (a - b)$, the wave-vectors is:

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

(1 mark)

For region $-b < x < 0$, the wave-vectors is:

$$\beta = \sqrt{\frac{2m(E + V_0)}{\hbar^2}}$$

(1 mark)

For region $0 < x < (a - b)$, the wave-function is:

$$\psi_I = Ae^{i\alpha x} + Be^{-i\alpha x}$$

(1 mark)

For region $-b < x < 0$, wave-vectors are:

$$\psi_{II} = Ce^{i\beta x} + De^{-i\beta x}$$

(1 mark)

Note that in the above, A – D are constants, and the wave-vectors and wave-functions are written generally, without restrictions on energy *E*.

(ii) To find the Bloch function $u(x)$ in each region, we only need to manipulate the wave-functions in (i) into the form $\psi(x) = u(x).\exp(ikx)$ [2 marks for seeing this].

$$\psi_I = Ae^{i\alpha x} + Be^{-i\alpha x} = e^{ikx}\left(Ae^{i(\alpha-k)x} + Be^{-i(\alpha+k)x}\right)$$
$$\Rightarrow u_I(x) = Ae^{i(\alpha-k)x} + Be^{-i(\alpha+k)x}$$

(2 marks)

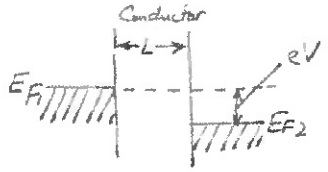
Similarly,

$$\psi_{II} = Ce^{i\beta x} + De^{-i\beta x} = e^{ikx}\left(Ce^{i(\beta-k)x} + De^{-i(\beta+k)x}\right)$$
$$\Rightarrow u_{II}(x) = Ce^{i(\beta-k)x} + De^{-i(\beta+k)x}$$

(2 marks)

Note that while the two Bloch functions are different due to the α and β terms, wave vector k is the same in both. The solution will need to indicate this, e.g. via the notation used.

(b)



Current I, across the conductor at applied voltage V is given by:

$$I = I_{right} - I_{left}$$

$$= e\int_0^\infty v(k)n(k)f(k)dk - e\int_0^\infty v(k')n(k')f(k')dk'$$

where $v(k)$ = electron velocity

$n(k) = 1/\pi$

$f(k)$ = Fermi-Dirac function [4]

Using $v = \frac{p}{m} = \frac{\hbar k}{m}$, [2]

$$I = e\int_0^\infty \frac{\hbar k}{m}\cdot\frac{1}{\pi}\cdot f(E)\frac{dk}{dE}dE - e\int_0^\infty \frac{\hbar k'}{m}\cdot\frac{1}{\pi}\cdot f(E')\frac{dk'}{dE'}dE'$$

$$= e\int_0^{E_{F_1}} \frac{\hbar k}{m}\frac{1}{\pi}\frac{dk}{dE}dE - e\int_0^{E_{F_2}} \frac{\hbar k'}{m}\cdot\frac{1}{\pi}\cdot\frac{dk'}{dE'}dE' \quad (\text{Assume } T=0K)$$ [2]

Furthermore, $E = \frac{p^2}{2m} = \frac{(\hbar k)^2}{2m} \Rightarrow \frac{dE}{dk} = \frac{\hbar^2 k}{m}$ [2]

$$\Rightarrow I = e\int_0^{E_{F_1}} \frac{\hbar k}{m}\cdot\frac{1}{\pi}\cdot\frac{m}{\hbar^2 k}dE - e\int_0^{E_{F_2}} \frac{\hbar k'}{m}\cdot\frac{1}{\pi}\cdot\frac{m}{\hbar^2 k'}dE'$$

$$= e\int_{E_{F_2}}^{E_{F_1}} \frac{1}{\hbar\pi}dE = \frac{e}{\hbar\pi}(E_{F_1} - E_{F_2})$$

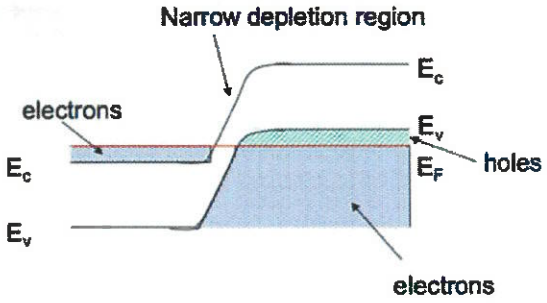
$$\Rightarrow I = \frac{e}{\frac{h}{2\pi}\cdot\pi}(eV) = \frac{2e^2}{h}\cdot V$$

$$\therefore G = \frac{I}{V} = \frac{2e^2}{h} \;//$$ [4]

(Total: 14 marks)

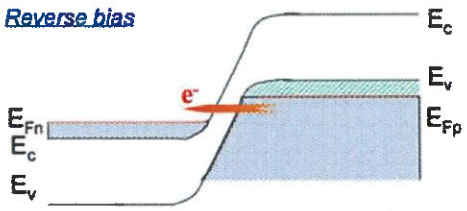
(c)

The Esaki tunnelling diode uses degenerately-doped n and p semiconductor regions to create conduction band – valance band tunnelling of electrons and holes across a narrow depletion region. Here, the heavy doping in the n and p regions restricts the depletion region width. The band diagram of the device at zero bias is shown below:



[1]

When the diode is reverse biased, electrons tunnel through the barrier formed by the energy gap in the depletion region, leading to a linear increase in current for reverse bias, electron flowing from the p (valance band) to n (conduction band) region:



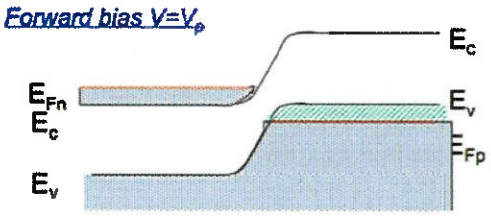
[1]

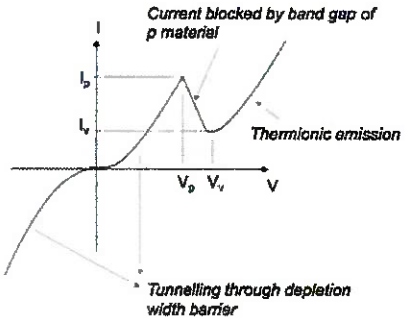
When the diode is forward biased, electrons tunnel from the conduction band of the n region to the valance band of the p region only as long as filled states in the conduction band overlap with empty states in the valance band (see the band diagram below, left).

[2]

As the bias increases further, the overlap ends and current falls, creating a current peak at V_p and a negative differential resistance region (see in the I-V characteristic below, right). Further increases in forward bias lead to increasing current flow due to electrons overcoming the n-p built-in potential barrier in the conduction band.

[2]


Forward bias V=Vp
Ec
EFn
Ec
Ev
Ev
EFp


Current blocked by band gap of p material
I
Ip
Iv
Thermionic emission
Vp Vv
V
Tunnelling through depletion width barrier

Answer, Question 4:

a) The main transduction mechanisms are:
Electrostatic (or capacitive)
Electromagnetic
Piezoresistive
Piezoelectric

Electrostatic and piezoresistive are the most widely used. The reason is that they can be implemented in a silicon mechanical layer without the need for any additional materials. Magnetic and piezoelectric transduction both require the introduction of materials that are difficult to process. Also, it is difficult to make efficient magnetic circuits using MEMS processes. [6]

b)i) The accelerating proof mass m_p can be treated as a point load of $P = -m_p a$ at a distance of $(l + L/2)$ from left-hand edge of the device. This load is shared equally between the two cantilevers, so the bending moment a distance x along either cantilever is:

$$M = -\frac{m_p}{2}(l + L/2 - x)a \qquad [4]$$

b)ii) The strain in a bent beam is $\varepsilon = -y/R$ where y is the distance from neutral axis and R is the radius of curvature. Combining this with the relation $M = EI/R$, and noting the fact that the piezoresistive bridge is at the top surface of the cantilever where $y = h/2$, we have:

$$\varepsilon = -\frac{h}{2}\frac{M}{EI} = \frac{h}{2}\frac{m_p(l + L/2)}{2EI}a$$

where we have evaluated M at the root of the cantilever, i.e. at $x = 0$. Using $m_p = WHL\rho$ and $I = wh^3/12$ this becomes:

$$\varepsilon = \frac{h}{2}\frac{WHL\rho(l + L/2)}{2E(wh^3/12)}a = \frac{3WHL(l + L/2)}{Ewh^2}\rho a \qquad [5]$$

b)iii) Considering the orientations of the piezoresistors, we see that R_1 and R_3 will have $\sigma_L = E\varepsilon, \sigma_T = 0$ while R_2 and R_4 will have $\sigma_T = E\varepsilon, \sigma_L = 0$. Assuming the resistors have the same nominal value R, their values will change to:

$$R_1 = R_3 = R(1 + \pi_L E\varepsilon) \;\; ; \;\; R_2 = R_4 = R(1 + \pi_T E\varepsilon)$$

The bridge output will then be:

$$V_{out} = \frac{R_1R_3 - R_2R_4}{(R_1 + R_2)(R_3 + R_4)}V_s = \frac{(1 + \pi_L E\varepsilon)^2 - (1 + \pi_T E\varepsilon)^2}{(2 + \pi_L E\varepsilon + \pi_T E\varepsilon)^2}V_s$$

Using the formula given in part b)ii), the stress $E\varepsilon$ at unit acceleration is 209.7 kPa, so we have $\pi_L E\varepsilon = 1.51 \times 10^{-4}$ and $\pi_T E\varepsilon = -1.38 \times 10^{-4}$. With V_s = 5 V, the above formula gives a sensitivity of 722 μV/(ms^{-2}). [5]

b)iv) The bending equation is $M = EIv'' = -\frac{m_p}{2}(l + L/2 - x)a$ where $v(x)$ is the deflection profile. Integrating once and applying the boundary condition $v'(0) = 0$, the slope is obtained as:

$$v'(x) = -\frac{m_p a}{2EI}[(l + L/2)x - x^2/2]$$

A second integration, with the boundary condition $v(0) = 0$, gives the deflection profile:

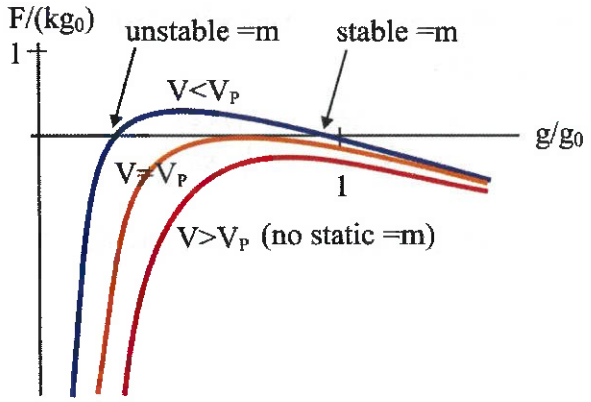
$$v(x) = -\frac{m_p a}{2EI}[(l + L/2)x^2/2 - x^3/6] \quad [6]$$

The deflection at the end of the proof mass is approximately $v_{end} = v(l) + v'(l)L$. The proof mass is $m_p = 2 \times 2 \times 0.4 \times 10^{-9} \times 2330 = 3.73 \times 10^{-6}$ kg, and $I = 2 \times 10^{-4} \times (20 \times 10^{-6})^3/12 = 1.33 \times 10^{-19}$ m^4. With E = 160 GPa, a = 100 ms^{-2}, $x = l$ = 500 μm and L = 2 mm, the above formulae give $v(l)$ = -1.46 μm and $v'(l) = -5.48 \times 10^{-3}$, so that v_{end} = -12.4 μm. [4]

Answer, Question 5:

a) The electrostatic force on the moveable plate is $F_e = -A\varepsilon_0 E^2/2$ where $E = V/g$ is the electric field in the gap. Here we are neglecting end effects and the dielectric. The only other force acting on the moveable plate is the spring force, $F_k = k(g_0 - g)$, so the total force is:

$$F = F_e + F_k = k(g_0 - g) - \frac{\varepsilon_0 A V^2}{2g^2} \qquad (1) \qquad [4]$$



The graph shows variation of force with gap for different values of V. When $V < V_P$ there are two equilibrium points (i.e. points where $F = 0$). One is stable, and the other is unstable, but the moveable plate naturally settles at the stable one (and never reaches the unstable one) if the applied voltage is increased from zero. For $V > V_P$ there is no equilibrium point, and the total force is always negative so the plate snaps down. [6]

b) The conditions at the point of snap-down are $F = 0$ and $\partial F/\partial g = 0$. Differentiating the force equation gives:

$$\frac{\partial F}{\partial g} = -k + \frac{\varepsilon_0 A V^2}{g^3} \qquad (2)$$

So, when $\partial F/\partial g = 0$ we have: $$\varepsilon_0 A V^2 = kg^3 \qquad (3)$$

Substituting (3) into (1), and setting $F = 0$, we find that $g = \frac{2g_0}{3}$ at the point of snap-down. Substituting this value of g into (3) gives the quoted result. [8]

c) The stiffness of each flexure is $12EI/L^3$, where L is the flexure length and $I = wt^3/12$ is the second moment of area, with w being the flexure width and t being the thickness of the gold mechanical layer. The hammock suspension comprises four flexures in parallel, so the total stiffness is $k = 48EI/L^3 = 4Ewt^3/L^3$. Putting $L = 250$ µm, $w = 4$ µm, $t = 2$ µm, $E = 80$ GPa we obtain $k = 0.655$ N/m.

With $k = 0.655$N/m, $g_0 = 4$ μm, and A = 72,500 μm^2, the snap-down voltage is obtained as V_P = 4.4 V. [6]

d) The initial electrostatic force is: $F_e = -\dfrac{A\varepsilon_0 V^2}{2g_0^2}$. This will cause the plate to accelerate at a rate of $a = F_e / m = -\dfrac{\varepsilon_0 V^2}{2g_0^2 \rho t}$ where we have used the relation $m = \rho A t$ for the mass of the plate. Assuming the force remains constant, the plate will move a distance $\frac{1}{2}at^2$ in time t, so the switching time T_s will be given by:

$$T_s = \sqrt{\frac{2g_0}{a}} = \frac{2}{V}\sqrt{\frac{\rho t g_0^3}{\varepsilon_0}}$$

Putting $\rho = 19{,}320$ kg/m^3, $t = 2$ μm, $g_0 = 4$ μm, V = 20 V, this gives $T_s = 53$ μs. [6]